Sonali Lonkar

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Professional Summary

Graduate Software Engineering student at San Jose State University with experience in developing scalable backend systems and full-stack applications, plus recent applied machine learning projects emphasizing reproducible training pipelines, time-series forecasting, and model benchmarking. Strong foundation in data structures, algorithms, and software engineering with hands-on expertise in Python, Java, C++, and modern cloud/devops tooling. Passionate about building reliable, high-performance products and data-driven systems that deliver measurable impact.

Technical Skills

- **Languages**: C++, Python, Java, JavaScript
- **Frameworks**: Spring Boot, Node.js, React, Angular
- **ML / DL**: PyTorch, PyTorch Lightning; Hugging Face Transformers (BERT/DistilBERT); TensorFlow/Keras; scikit-learn; XGBoost
- **Time Series / Forecasting**: Walk-forward validation; multi-horizon forecasting; HAR/HARX; GARCH; GRU/LSTM; TFT, Informer, N-BEATS
- **Data & Evaluation**: pandas, numpy, scipy; StandardScaler; PR-AUC/F1; calibration; statistical significance testing; backtesting with transaction costs
- **Pipelines & Tools**: YAML configs (PyYAML), Makefile, shell scripting, pytest, Jupyter
- **Backend / Deployment**: Flask (REST APIs), Streamlit demos; Docker, Jenkins, GitLab CI/CD; AWS
- **Databases & Systems**: MySQL, MongoDB, Linux, Git, Perforce, JIRA, Postman, Maven

Education

San Jose State University – M.S. in Software Engineering | Expected Dec 2026 | GPA: 3.6
CDAC Mumbai – Post Graduate Diploma in Advanced Computing
Government College of Engineering, Amravati – B.Tech in Electrical Engineering

Professional Experience

Software Developer – Agnie Media Software, Pune, India | Jan 2021 – Mar 2022

- Designed and deployed Java Spring Boot microservices supporting high-performance REST APIs.
- Automated CI/CD pipelines using Jenkins and Docker, reducing deployment time by 40%.
- Collaborated with QA and infrastructure teams for system debugging and resource optimization.
- Built and integrated backend APIs with MySQL databases for financial services applications.
- Conducted debugging and optimization to improve API response times by 30%.

Academic Projects

Deep Learning - Volatility Forecasting | December 2025

- Implemented multi-horizon (1/5/22 day) volatility forecasting with TFT, Informer, N-BEATS, GRU/LSTM and MLP baselines using PyTorch Lightning.
- Benchmarked deep models against statistical baselines (HAR/HARX via statsmodels; GARCH via arch) and gradient-boosted trees (XGBoost) with walk-forward validation.
- Automated batch experiments using YAML configs (PyYAML), Makefile targets, and shell scripts; stored datasets efficiently in Parquet (pyarrow).
- Evaluated strategy robustness via portfolio backtesting with transaction costs and statistical significance testing; generated plots/reports for model comparisons.

Toxic Comment Classification: Interpretable Benchmark for Data Mining | December 2025

- Built multi-label toxicity classification across 6 dimensions; compared DistilBERT (Transformers + PyTorch) vs TF-IDF baselines (LogReg/SVM/RF/XGBoost) and LSTM variants.

- Used temporal splits, hyperparameter search, and comprehensive metrics (PR-AUC, F1, calibration); added interpretability with SHAP and extended evaluation (out-of-domain tests).
- Delivered a production-style pipeline with a Flask REST API and Streamlit demo; automated dataset acquisition via Kaggle API and scripted training/evaluation runs.

Predicting Historical Stock Prices With Machine Learning | December 2025
- Engineered sliding-window features (2–5 day windows) and trained classical ML regressors (Linear/Ridge/Poly, RF/GB, MLP, XGBoost) plus an LSTM regressor in PyTorch.
- Implemented 5-fold walk-forward cross-validation with feature normalization (StandardScaler) and automated data prep/training via Makefile and shell scripts.
- Aggregated results into reproducible reports to support benchmarking and model selection.

Supermemory Assistant - AI-Powered Personal Assistant | December 2025
- Built a full-stack AI assistant with dynamic personas and persistent long-term memory using Flask and Next.js.
- Integrated Gemini API for intelligent responses and Supermemory API for memory storage, classification, and contextual recall. Also added file uploads with OCR, web search, calendar integration, and automated workflows via external service connectors.

Campus peer-to-peer marketplace MVP for Software Systems | December 2025
- Designed and built a campus peer-to-peer marketplace with a React frontend and Spring Boot 3 (Java 17) backend; modeling a normalized MySQL schema, deploying the DB on Amazon RDS and APIs on EC2, and using Flyway, Swagger, Docker, and Postman for rapid iteration.

ReliefCircle – Cloud-Native Donation & Charity Management Platform | May 2025
- Designed a scalable distributed platform using Java Spring Boot, MySQL, and AWS.
- Implemented role-based authentication, payment integration, and real-time notifications (AWS SNS).

Distributed Twitter Clone – Scalable Microservice-Based System | April 2025
- Built a distributed social platform using Flask, MongoDB, and REST APIs.
- Implemented data sharding, load balancing, and fault-tolerant microservices for high availability.